
Solving Linear Systems of First-Order ODEs via the Inverse Function Theorem

Sammith R Belur

MA 442: Honors Real Analysis II - Final Project
Spring 2026

September 24, 2026

CONTENTS

1. Introduction	2
2. Classical Methods to solve Linear Systems of First-Order ODEs	3
2.1. The Matrix Exponential	3
2.2. Diagonalization and Jordan Forms	4
2.3. The Fundamental Matrix	5
2.4. Non-Autonomous Case Extension	5
3. Relevant Theorems and Definitions	7
3.1. Normed Vector Spaces and Banach Spaces	7
3.2. Bounded Linear Maps	8
3.3. The Banach Fixed Point Theorem	8
3.4. The Open Mapping Theorem	9
3.5. The Fréchet Derivative	11
3.6. Gronwall's Inequality	11
4. The Inverse Function Theorem (IFT)	13
4.1. Single Variable Case	13
4.2. The $\mathbb{R}^n$ Case	13
4.3. Generalization to Banach Spaces	14
5. How IFT is used to solve Linear Systems of First-Order ODEs	15
5.1. Reformulation of the problem	15
5.2. Analysis on the operator	15
5.3. The Key Idea	16
5.4. The Nonlinear Case	17
6. Extension to Non-Autonomous Systems	18
7. Acknowledgements	19
References	20

1. INTRODUCTION

Systems of ordinary differential equations (ODEs) arise in the study of mathematics and also across several applications [BDP]. Rather than keeping track of a single quantity over time, a system of ODEs tracks several quantities simultaneously and describes how they influence one another. They appear in classical mechanics, circuit analysis, population dynamics, and control theory when we want to analyze a system near its equilibrium points to name a few applications.

A key distinction in the study of such systems is between linear and nonlinear systems. Typically, a problem for a system of ODEs takes the form

$$x'(t) = f(t, x(t)), \quad x(t_0) = x_0,$$

where $x(t) \in \mathbb{R}^n$ and $f : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$. In the linear case, the right-hand side depends on x only through a constant matrix $A \in \mathbb{R}^{n \times n}$, giving the system $x'(t) = Ax(t)$. A solution to this system, together with an initial condition $x(0) = x_0$, is a differentiable function $x : [0, T] \rightarrow \mathbb{R}^n$ satisfying both conditions simultaneously.

One application of a system of a linear first order ODE is a pair of coupled oscillators [BDP], where we take

$$x'_1 = -x_1 + x_2, \quad x'_2 = x_1 - x_2,$$

which can be written as $x' = Ax$ with $A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$ and $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ with $x' = \begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix}$

In the nonlinear case, $f(t, x(t))$ may depend on x in more complex ways. A classical example is the Lotka–Volterra predator-prey system [M]

$$u' = \alpha u - \beta uv, \quad v' = \delta uv - \gamma v,$$

where u and v represent prey and predator populations respectively, and $\alpha, \beta, \gamma, \delta > 0$ are constants. The quadratic interaction terms uv make this system nonlinear.

A further distinction is between autonomous and non-autonomous systems. A system is called *autonomous* if f does not depend explicitly on t , so the equation takes the form $x'(t) = f(x(t))$. Otherwise, when f depends explicitly on time, the system is *non-autonomous*. In the linear setting, the autonomous case corresponds to $x'(t) = Ax(t)$ with A a constant matrix, while the non-autonomous case allows the matrix to vary with time, giving $x'(t) = A(t)x(t)$. [BDP]

Both linear and nonlinear systems have been studied extensively, and several methods exist for finding their solutions. In this report we focus primarily on linear systems, for which explicit solution methods are well developed and will be discussed in detail in Section 2. The linear autonomous case $x'(t) = Ax(t)$ is the simplest nontrivial setting in which the ideas of this report can be developed cleanly, and the framework extends naturally from there to non-autonomous and nonlinear systems. Nonlinear systems appear later in Section 5.4, where we compare and contrast how approaches work for linear and nonlinear systems.

The primary goal of this report is to examine the linear autonomous system

$$x'(t) = Ax(t), \quad x(0) = x_0,$$

through the lens of the Inverse Function Theorem. Rather than developing a new solution method, we give a detailed account of how the Inverse Function Theorem gives us a rigorous framework for understanding why a solution exists, why it is unique, and why it depends smoothly on the initial data x_0 . This is covered in more detail in Section 5.

2. CLASSICAL METHODS TO SOLVE LINEAR SYSTEMS OF FIRST-ORDER ODES

In this section we discuss three main approaches to find the solution to a linear system of first order ODEs: matrix exponential, diagonalization, and the fundamental matrix.

2.1. The Matrix Exponential. The matrix exponential provides the most direct route to solving the linear system

$$x'(t) = Ax(t), \quad x(0) = x_0.$$

Definition 2.1. For $A \in \mathbb{R}^{n \times n}$ and $t \in \mathbb{R}$, the *matrix exponential* is defined by the power series

$$e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = I + At + \frac{(At)^2}{2!} + \dots$$

This series converges absolutely for all $t \in \mathbb{R}$ and all $A \in \mathbb{R}^{n \times n}$

The matrix exponential satisfies several important properties [Y; M]:

- (i) $e^{A \cdot 0} = I$;
- (ii) $\frac{d}{dt} e^{At} = Ae^{At} = e^{At} A$;
- (iii) $e^{At} e^{As} = e^{A(t+s)}$;
- (iv) e^{At} is invertible for all t , with $(e^{At})^{-1} = e^{-At}$;
- (v) If $AB = BA$, then $e^{At} e^{Bt} = e^{(A+B)t}$.

Property (ii) immediately gives us that $x(t) = e^{At}x_0$ solves $x'(t) = Ax(t)$ with $x(0) = x_0$.

Theorem 2.2. Let $A \in \mathbb{R}^{n \times n}$ and $x_0 \in \mathbb{R}^n$. Then $x(t) = e^{At}x_0$ is the unique solution to the initial value problem $x'(t) = Ax(t)$, $x(0) = x_0$.

Proof. That $x(t) = e^{At}x_0$ is a solution follows directly from property (ii). Uniqueness is established in Section 3.6 as a consequence of the Gronwall inequality, once the necessary tools are in place. $\square$

Example 2.3. Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ and $x_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. We write $A = D + N$ where $D = 2I$ and $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Since $D = 2I$ commutes with every matrix and $N^2 = 0$, property (v) gives

$$e^{At} = e^{Dt} e^{Nt} = e^{2t}(I + Nt) = e^{2t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

The solution is therefore

$$x(t) = e^{At}x_0 = \begin{pmatrix} e^{2t} + te^{2t} \\ e^{2t} \end{pmatrix}.$$

Limitation. While an elegant method, computing e^{At} in closed form is often very difficult. The approach used in the example above was rather simple because of a special structure of A . For a general matrix, we have to find solutions via diagonalization or the Jordan normal form, which we discuss next.

2.2. Diagonalization and Jordan Forms. When A has n linearly independent eigenvectors, the system $x'(t) = Ax(t)$ can be decoupled into n independent scalar equations, each of which is straightforward to solve. We do this via diagonalization.

We assume that A has eigenvalues $\lambda_1, \dots, \lambda_n$ with corresponding linearly independent eigenvectors $v_1, \dots, v_n$.

Using the eigenvectors, we form the matrix $P = [v_1 \mid \dots \mid v_n]$ and the diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, so that $A = PDP^{-1}$. The substitution $x(t) = Py(t)$ transforms the system into

$$Py'(t) = APy(t) = PDP^{-1}Py(t) = PDy(t),$$

and since P is invertible, this gives $y'(t) = Dy(t)$. This decouples into n independent scalar equations

$$y_i'(t) = \lambda_i y_i(t), \quad i = 1, \dots, n,$$

each solved by $y_i(t) = e^{\lambda_i t} y_i(0)$, which is the classical solution to first order ODES [BDP].

The solution to the original system is then recovered via $x(t) = Py(t)$

Example 2.4. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and $x_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. The eigenvalues of A are $\lambda_1 = 3$ and $\lambda_2 = -1$, with corresponding eigenvectors $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. So we have

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}, \quad P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The initial condition transforms to $y(0) = P^{-1}x_0 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, giving

$$y(t) = \frac{1}{2} \begin{pmatrix} e^{3t} \\ e^{-t} \end{pmatrix}.$$

Recovering $x(t) = Py(t)$,

$$x(t) = \frac{1}{2} \begin{pmatrix} e^{3t} + e^{-t} \\ e^{3t} - e^{-t} \end{pmatrix}.$$

Limitation. Diagonalization requires A to have a full set of n linearly independent eigenvectors (known as an eigenbasis). When this fails - that is, when A is *defective* - we have

to instead use the Jordan normal form. In the Jordan form, $A = PJP^{-1}$ where J is block diagonal with Jordan blocks of the form

$$J_k = \begin{pmatrix} \lambda_k & 1 & & \\ & \lambda_k & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_k \end{pmatrix}.$$

As a brief example, let us look at $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, which has the repeated eigenvalue $\lambda = 2$ but only one linearly independent eigenvector. Here $A = 2I + N$ where $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is nilpotent with $N^2 = 0$, and A is already in Jordan form. The matrix exponential then follows as in the previous subsection. In general, computing the Jordan form and the exponential becomes more difficult and computationally expensive as the size of A grows [Y].

2.3. The Fundamental Matrix. The fundamental matrix provides another framework for solving linear systems that works regardless of whether A is diagonalizable, and extends naturally to the non-autonomous case.

Definition 2.5. [Y] The *fundamental matrix* $\Phi(t)$ for the system $x'(t) = A(t)x(t)$ is the $n \times n$ matrix-valued function satisfying

$$\Phi'(t) = A(t)\Phi(t), \quad \Phi(0) = I.$$

Once $\Phi(t)$ is known, the solution to $x'(t) = Ax(t)$, $x(0) = x_0$ is simply

$$x(t) = \Phi(t)x_0.$$

A key advantage of this approach is that $\Phi(t)$ is computed once and then the solution for any initial condition x_0 is recovered immediately by matrix multiplication, without repeating the computation.

Theorem 2.6 (Abel's Theorem [M]). *Let $\Phi(t)$ be the fundamental matrix for $x'(t) = A(t)x(t)$. Then*

$$\det \Phi(t) = \exp\left(\int_0^t \text{tr}(A(s)) ds\right) \neq 0$$

for all t . In particular, $\Phi(t)$ is invertible for all t .

In the autonomous case $A(t) = A$, this reduces to $\det \Phi(t) = e^{t \cdot \text{tr}(A)} \neq 0$, and the fundamental matrix coincides with the matrix exponential: $\Phi(t) = e^{At}$.

2.4. Non-Autonomous Case Extension. The fundamental matrix framework extends naturally to the non-autonomous forced system

$$x'(t) = A(t)x(t) + h(t), \quad x(0) = x_0.$$

Following [Y], we use $\Phi(t)$ as an integrating factor. Writing $x(t) = \Phi(t)u(t)$ and substituting,

$$\Phi(t)u'(t) = h(t) \implies u'(t) = \Phi(t)^{-1}h(t)$$

Integrating and using $u(0) = x_0$ gives

$$u(t) = x_0 + \int_0^t \Phi(s)^{-1} h(s) ds,$$

and so the solution is

$$(1) \quad x(t) = \Phi(t)x_0 + \int_0^t \Phi(t)\Phi(s)^{-1}h(s) ds.$$

This is the *variation of parameters formula*. We also introduce the *propagator* $\Phi(t, s) = \Phi(t)\Phi(s)^{-1}$ for $0 \leq s \leq t$, which satisfies

$$\frac{\partial}{\partial t}\Phi(t, s) = A(t)\Phi(t, s), \quad \Phi(s, s) = I,$$

so that (1) can be written compactly as

$$x(t) = \Phi(t, 0)x_0 + \int_0^t \Phi(t, s)h(s) ds.$$

Limitation. The fundamental matrix provides a conceptual unification but not a computational shortcut. Computing $\Phi(t)$ explicitly still requires solving $\Phi'(t) = A\Phi(t)$, which itself requires the matrix exponential or diagonalization. The framework reframes the problem but still requires some amount of computation for large systems.

Each of the methods we analyzed address the question of how to compute a solution for this system of ODEs. In the sections that come up, we look at tools that we need to understand when and why a solution should exist at all.

3. RELEVANT THEOREMS AND DEFINITIONS

Before we build up to the Inverse Function Theorem and apply it to systems of ODEs, we will first establish the analytical setting in which this argument lives. We state definitions, theorems and write relevant proofs that we covered in class for MA 442, but it is still included here to serve as a self-contained reference before the core of the report.

3.1. Normed Vector Spaces and Banach Spaces. We begin with the notion of a norm, which gives us a way to measure the size of elements in a vector space.

Definition 3.1 (Definition 113, [D]). Let V be a vector space. A function $\|\cdot\| : V \rightarrow [0, \infty)$ is a *semi-norm* if for all $\alpha \in \mathbb{R}$ and $x, y \in V$:

- (i) $\|\alpha x\| = |\alpha| \|x\|$;
- (ii) $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

If additionally $\|x\| > 0$ for all $x \neq 0_V$, then $\|\cdot\|$ is called a *norm*, and $(V, \|\cdot\|)$ is called a *normed vector space*.

Intuitively, $\|x\|$ measures how large x is. Every norm induces a metric $d(x, y) = \|x - y\|$, so a normed vector space is automatically a metric space. This allows us to talk about convergence, continuity, and completeness in V .

Example 3.2. The space $\mathbb{R}^n$ equipped with the Euclidean norm $\|x\|_2 = (\sum_{i=1}^n x_i^2)^{1/2}$ is a normed vector space. The space $C([0, T], \mathbb{R}^n)$ of continuous functions $f : [0, T] \rightarrow \mathbb{R}^n$, equipped with the sup-norm

$$\|f\|_\infty = \sup_{t \in [0, T]} |f(t)|,$$

is also a normed vector space. This is the function space that will appear naturally when we study the solutions of ODEs

A normed vector space gives us a way to measure size, but not yet a guarantee that limiting operations stay within the space. For that we need completeness.

Definition 3.3 (Definitions 56, 57, [D]). Let X be a normed vector space. A sequence $\{x_k\}$ in X is called a *Cauchy sequence* if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\|x_m - x_n\| < \varepsilon \quad \text{whenever } m, n \geq N.$$

Definition 3.4 (Definition 122, [D]). A normed vector space $(V, \|\cdot\|)$ is called a *Banach space* if it is complete with respect to the metric arising from $\|\cdot\|$ — that is, every Cauchy sequence in V converges to a limit in V .

Completeness is essential throughout this report. It is what makes fixed point arguments work, and it is assumed by both the Banach Fixed Point Theorem and the Open Mapping Theorem that follow.

Example 3.5. The space $\mathbb{R}^n$ with the Euclidean norm is a Banach space, as a consequence of the completeness of $\mathbb{R}$. The space $C([0, T], \mathbb{R}^n)$ with the supremum norm is also a Banach space: a Cauchy sequence of continuous functions converges uniformly, and the uniform limit of continuous functions is again continuous.

The two function spaces central to our analysis are:

$$\begin{aligned} X &= C^1([0, T], \mathbb{R}^n), & \|x\|_{C^1} &= \|x\|_\infty + \|x'\|_\infty, \\ Y &= C([0, T], \mathbb{R}^n) \times \mathbb{R}^n, & \|(f, v)\|_Y &= \|f\|_\infty + |v|. \end{aligned}$$

Both X and Y are Banach spaces. For X , a Cauchy sequence in the C^1 norm forces both the functions and their derivatives to converge uniformly, and the limit is again C^1 . The space Y is a product of two Banach spaces, which is itself a Banach space under the given norm.

3.2. Bounded Linear Maps. We have now seen Banach Spaces, and consider maps between them. Here, we work with bounded linear maps, which are continuous linear functions between normed vector spaces.

Definition 3.6 (Definition 95, [D]). Given normed vector spaces X and Y , a function $L : X \rightarrow Y$ is called *linear* if

$$L(\alpha x + \beta y) = \alpha L(x) + \beta L(y)$$

for all $\alpha, \beta \in \mathbb{R}$ and all $x, y \in X$.

Definition 3.7. A linear map $L : X \rightarrow Y$ between normed vector spaces is *bounded* if there exists a constant $C \geq 0$ such that

$$\|Lx\|_Y \leq C \|x\|_X \quad \text{for all } x \in X.$$

The *operator norm* of L is defined to be

$$\|L\|_{\text{op}} = \sup_{\|x\|_X=1} \|Lx\|_Y.$$

A linear map between normed spaces is bounded if and only if it is continuous — boundedness and continuity are equivalent for linear maps, unlike for general functions. This means the operator norm $\|L\|_{\text{op}}$ is the smallest constant C for which the bound $\|Lx\|_Y \leq C \|x\|_X$ holds.

Example 3.8. Any matrix $A \in \mathbb{R}^{n \times n}$ defines a bounded linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $L(x) = Ax$. The operator norm in this case is the largest singular value of A , which is finite since A is a finite matrix.

Definition 3.9 (Definition 97, [D]). A linear map $L : X \rightarrow Y$ is called a *linear isomorphism* if it is bijective. If L is additionally bounded and its inverse $L^{-1} : Y \rightarrow X$ is also bounded, we call L a *bounded linear isomorphism*.

The space of all bounded linear maps from X to Y is denoted $\mathcal{L}(X, Y)$, and is a Banach space under the operator norm when Y is a Banach space.

3.3. The Banach Fixed Point Theorem. The Banach Fixed Point Theorem is the most important tool for the proof of the Inverse Function Theorem. It guarantees that certain maps have a unique fixed point, and gives a constructive way to find it.

Definition 3.10. Let (X, d) be a metric space. A map $T : X \rightarrow X$ is called a *contraction* if there exists a constant $c \in [0, 1)$ such that

$$d(T(x), T(y)) \leq c d(x, y) \quad \text{for all } x, y \in X.$$

A point $x^* \in X$ is called a *fixed point* of T if $T(x^*) = x^*$.

Theorem 3.11 (Banach Fixed Point Theorem [TTH]). *Let (X, d) be a complete metric space and let $T : X \rightarrow X$ be a contraction with constant $c \in [0, 1)$. Then T has a unique fixed point $x^* \in X$. Moreover, for any starting point $x_0 \in X$, the iterates $x_{n+1} = T(x_n)$ converge to x^* , with the error bound*

$$d(x_n, x^*) \leq \frac{c^n}{1-c} d(x_1, x_0).$$

Proof. Existence. Pick any $x_0 \in X$ and define the sequence $x_{n+1} = T(x_n)$. We first show this is a Cauchy sequence. Since T is a contraction,

$$d(x_{n+1}, x_n) = d(T(x_n), T(x_{n-1})) \leq c d(x_n, x_{n-1}) \leq \cdots \leq c^n d(x_1, x_0).$$

For $m > n$, the triangle inequality gives

$$\begin{aligned} d(x_m, x_n) &\leq d(x_m, x_{m-1}) + \cdots + d(x_{n+1}, x_n) \\ &\leq (c^{m-1} + \cdots + c^n) d(x_1, x_0) \\ &= c^n \cdot \frac{1 - c^{m-n}}{1 - c} d(x_1, x_0) \\ &\leq \frac{c^n}{1 - c} d(x_1, x_0). \end{aligned}$$

Since $c < 1$, the right-hand side tends to 0 as $n \rightarrow \infty$, so $\{x_n\}$ is Cauchy. Since X is complete, there exists $x^* \in X$ with $x_n \rightarrow x^*$.

x^* is a fixed point. Since T is a contraction it is continuous, so

$$T(x^*) = T\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} T(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x^*.$$

Uniqueness. Suppose y^* is another fixed point. Then

$$d(x^*, y^*) = d(T(x^*), T(y^*)) \leq c d(x^*, y^*).$$

Since $c < 1$, this forces $d(x^*, y^*) = 0$, so $x^* = y^*$.

Error bound. Taking $m \rightarrow \infty$ in the Cauchy estimate gives $d(x_n, x^*) \leq \frac{c^n}{1-c} d(x_1, x_0)$. $\square$

The completeness of X is essential here — without it, the Cauchy sequence $\{x_n\}$ might not converge to anything within the space. This is why we need to work in Banach spaces.

3.4. The Open Mapping Theorem. The Open Mapping Theorem is what allows us to work with inverses and show that they are bounded, without having to compute it explicitly. We build up to it through a key lemma.

Definition 3.12 (Definition 16, [D]). Given a function $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ between topological spaces, we say f is an *open mapping* if for each open set $V \in \mathcal{T}_X$, we have $f(V) \in \mathcal{T}_Y$.

Lemma 3.13 ([TTII]). *Let X be a Banach space and let $L : X \rightarrow X$ be a bounded linear map with $\|L\|_{\text{op}} < 1$. Then $I - L$ is a bounded linear isomorphism, and*

$$(I - L)^{-1} = \sum_{k=0}^{\infty} L^k,$$

where the series converges in the operator norm.

Proof. Since $\|L\|_{\text{op}} < 1$, the series $\sum_{k=0}^{\infty} L^k$ converges absolutely in $\mathcal{L}(X, X)$:

$$\sum_{k=0}^{\infty} \|L^k\|_{\text{op}} \leq \sum_{k=0}^{\infty} \|L\|_{\text{op}}^k = \frac{1}{1 - \|L\|_{\text{op}}} < \infty.$$

Since $\mathcal{L}(X, X)$ is a Banach space, the series converges to a bounded linear map $S = \sum_{k=0}^{\infty} L^k$. A direct computation shows

$$(I - L)S = S(I - L) = \sum_{k=0}^{\infty} L^k - \sum_{k=1}^{\infty} L^k = I,$$

so $S = (I - L)^{-1}$ and $I - L$ is a bounded linear isomorphism. $\square$

Theorem 3.14 (Open Mapping Theorem [TTII]). *Let X and Y be Banach spaces and let $L : X \rightarrow Y$ be a bounded linear bijection. Then L is an open mapping, and in particular $L^{-1} : Y \rightarrow X$ is bounded.*

Proof. It suffices to show that L maps the open unit ball $B_X = \{x \in X : \|x\|_X < 1\}$ to a set containing an open ball around 0 in Y .

Step 1: The closure $\overline{L(B_X)}$ contains a ball. Since L is surjective,

$$Y = \bigcup_{n=1}^{\infty} L(nB_X) = \bigcup_{n=1}^{\infty} n \overline{L(B_X)}.$$

By the Baire Category Theorem, Y cannot be written as a countable union of nowhere dense sets, so at least one $n \overline{L(B_X)}$ has nonempty interior. Hence $\overline{L(B_X)}$ itself contains an open ball, say $B_Y(y_0, r) \subset \overline{L(B_X)}$ for some y_0 and $r > 0$. Since $\overline{L(B_X)}$ is symmetric (as L is linear and B_X is symmetric), $B_Y(0, r) \subset \overline{L(B_X)}$.

Step 2: $L(B_X)$ contains a ball. We show $B_Y(0, r/2) \subset L(B_X)$. Let $y \in Y$ with $\|y\|_Y < r/2$. Since $B_Y(0, r) \subset \overline{L(B_X)}$, we can find $x_1 \in B_X$ with $\|y - Lx_1\|_Y < r/2$. Now $y - Lx_1$ satisfies $\|y - Lx_1\|_Y < r/2$, so scaling gives $2(y - Lx_1)/r \in \overline{L(B_X)}$, and we can find x_2 with $\|x_2\|_X < 1/2$ and $\|y - Lx_1 - Lx_2\|_Y < r/4$. Continuing this process gives a sequence $\{x_k\}$ with $\|x_k\|_X < 2^{-(k-1)}$ and $\|y - L(x_1 + \cdots + x_k)\|_Y < r \cdot 2^{-k}$. Setting $x = \sum_{k=1}^{\infty} x_k$, which converges in X since X is complete and $\sum \|x_k\|_X < 2$, we get $Lx = y$ and $\|x\|_X < 2$. Since $\|y\|_Y < r/2$, replacing y by $(r/2)^{-1}y$ and scaling shows that every y with $\|y\|_Y < r/4$ has a preimage with $\|x\|_X < 1$, so $B_Y(0, r/4) \subset L(B_X)$.

Since $L(B_X)$ contains an open ball around 0 in Y , and L is linear, L maps every open set to an open set. In particular, L^{-1} is continuous, which also gives us that it is bounded $\square$

Remark 3.15. The key consequence we use throughout this report is simple: if $F : X \rightarrow Y$ is a bounded linear bijection between Banach spaces, then F^{-1} is automatically bounded.

3.5. The Fréchet Derivative. The classical derivative from real analysis applies to functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. However, we work with maps between more general Banach spaces, so we need a more general notion of derivative. The Fréchet derivative is the correct generalization - it is the best bounded linear approximation to a map near a given point, in exactly the same spirit as the classical derivative.

Definition 3.16 ([VS]). Let V and W be normed vector spaces, $U \subseteq V$ open, and $f : U \rightarrow W$. We say f is *Fréchet differentiable at $x \in U$* if there exists a continuous linear function $L_{f,x} : V \rightarrow W$ such that

$$\lim_{\|v\|_V \rightarrow 0} \frac{\|f(x+v) - f(x) - L_{f,x}(v)\|_W}{\|v\|_V} = 0.$$

The map $L_{f,x}$ is unique and is called the *Fréchet derivative of f at x* , denoted $Df(x)$.

In other words, $Df(x)$ is the unique bounded linear map such that the remainder $f(x+v) - f(x) - Df(x)v$ is $o(\|v\|_V)$ as $\|v\|_V \rightarrow 0$. When $V = W = \mathbb{R}$, this reduces exactly to the classical derivative: $Df(x)$ is multiplication by $f'(x)$. When $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$, the Fréchet derivative coincides with the Jacobian matrix [VS].

Definition 3.17 (Definition 155, [D]). We say f is *continuously Fréchet differentiable on U* , or C^1 on U , if f is Fréchet differentiable at each $x \in U$ and the map $Df : U \rightarrow \mathcal{L}(V, W)$ is continuous with respect to the operator norm on $\mathcal{L}(V, W)$.

The following proposition is central to the main argument of this report. It says that a linear map is its own Fréchet derivative everywhere.

Proposition 3.18. [VS] *Let X and Y be Banach spaces and let $F : X \rightarrow Y$ be a bounded linear map. Then F is Fréchet differentiable at every $x \in X$, with $DF(x) = F$.*

Proof. Fix any $x \in X$ and consider the remainder when we perturb by $h \in X$:

$$F(x+h) - F(x) - F(h) = 0$$

by linearity of F . The remainder is identically zero for all h , so trivially

$$\lim_{\|h\|_X \rightarrow 0} \frac{\|F(x+h) - F(x) - F(h)\|_Y}{\|h\|_X} = 0.$$

Since F is bounded and linear, the definition of the Fréchet derivative is satisfied with $L_{F,x} = F$. Since x was arbitrary, $DF(x) = F$ for all $x \in X$. $\square$

3.6. Gronwall's Inequality. Gronwall's inequality is a fundamental tool in ODE theory. It converts an integral inequality into an explicit exponential bound, and will be used in Section 5 to establish injectivity of our operator F , and again to derive continuous dependence of solutions on initial data.

Lemma 3.19 (Gronwall's Inequality, [Y]). *Let $u, \beta : [0, T] \rightarrow [0, \infty)$ be continuous and let $\alpha_0 \geq 0$ be a constant. Suppose that*

$$u(t) \leq \alpha_0 + \int_0^t \beta(s) u(s) ds \quad \text{for all } t \in [0, T].$$

Then

$$u(t) \leq \alpha_0 \exp\left(\int_0^t \beta(s) ds\right) \quad \text{for all } t \in [0, T].$$

Proof. Define $v(t) = \int_0^t \beta(s)u(s) ds$, so that $v(0) = 0$ and $v'(t) = \beta(t)u(t)$. The hypothesis gives $u(t) \leq \alpha_0 + v(t)$, so

$$v'(t) = \beta(t)u(t) \leq \beta(t)(\alpha_0 + v(t)) = \alpha_0\beta(t) + \beta(t)v(t).$$

Let $B(t) = \int_0^t \beta(s) ds$. Multiplying both sides by the integrating factor $e^{-B(t)}$,

$$\frac{d}{dt}[e^{-B(t)}v(t)] = e^{-B(t)}(v'(t) - \beta(t)v(t)) \leq \alpha_0\beta(t)e^{-B(t)}.$$

Integrating from 0 to t and using $v(0) = 0$,

$$e^{-B(t)}v(t) \leq \alpha_0 \int_0^t \beta(s)e^{-B(s)} ds = \alpha_0(1 - e^{-B(t)}).$$

Hence $v(t) \leq \alpha_0(e^{B(t)} - 1)$, and so

$$u(t) \leq \alpha_0 + v(t) \leq \alpha_0 e^{B(t)} = \alpha_0 \exp\left(\int_0^t \beta(s) ds\right).$$

□

Remark 3.20. The special case $\beta(s) = \|A\|$ constant gives

$$u(t) \leq \alpha_0 e^{\|A\|t},$$

which is the form we will use in Section 5 when working with the system $x' = Ax$.

4. THE INVERSE FUNCTION THEOREM (IFT)

The Inverse Function Theorem is the core of this report. It basically says that if a function has an invertible derivative at a point, then the function itself is locally invertible near that point. We build up from the single variable case to the $\mathbb{R}^n$ case, and then briefly describe how the result generalizes to Banach spaces.

4.1. Single Variable Case.

Theorem 4.1 (Inverse Function Theorem, [Tao1]). *Let $X, Y \subseteq \mathbb{R}$ and let $f : X \rightarrow Y$ be invertible with inverse $f^{-1} : Y \rightarrow X$. Suppose $x_0 \in X$ and $y_0 = f(x_0) \in Y$ are limit points. If*

- (i) f is differentiable at x_0 ,
- (ii) f^{-1} is continuous at y_0 , and
- (iii) $f'(x_0) \neq 0$,

then f^{-1} is differentiable at y_0 and

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

Proof. We want to compute the derivative of f^{-1} at y_0 from the definition. Consider the difference quotient for f^{-1} at y_0 :

$$\frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0}.$$

Substitute $x = f^{-1}(y)$, so that $y = f(x)$ and $y_0 = f(x_0)$. The difference quotient becomes

$$\frac{x - x_0}{f(x) - f(x_0)}.$$

Since f^{-1} is continuous at y_0 , as $y \rightarrow y_0$ we have $x \rightarrow x_0$. Taking the limit and using $f'(x_0) \neq 0$,

$$(f^{-1})'(y_0) = \lim_{y \rightarrow y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \lim_{x \rightarrow x_0} \frac{x - x_0}{f(x) - f(x_0)} = \frac{1}{f'(x_0)}. \quad \square$$

4.2. The $\mathbb{R}^n$ Case. We now generalize to functions between open subsets of $\mathbb{R}^n$. The condition $f'(x_0) \neq 0$ from the single variable case is replaced by invertibility of the Jacobian matrix $Df(x_0)$, which is its natural generalization to higher dimensions.

Theorem 4.2 (Inverse Function Theorem, [Tao2]). *Let $E \subseteq \mathbb{R}^n$ be open and let $f : E \rightarrow \mathbb{R}^n$ be continuously differentiable on E . Suppose $x_0 \in E$ is such that the Jacobian $Df(x_0) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible. Then there exist open sets $U \subset E$ and $V \subset \mathbb{R}^n$ with $x_0 \in U$ and $f(x_0) \in V$ such that $f : U \rightarrow V$ is a bijection. The inverse $f^{-1} : V \rightarrow U$ is continuously differentiable, and*

$$D(f^{-1})(f(x_0)) = (Df(x_0))^{-1}.$$

Proof. Without loss of generality, assume $x_0 = 0$, $f(0) = 0$, and $Df(0) = I$. This can always be arranged by first shifting: replacing $f(x)$ with $f(x + x_0) - f(x_0)$ moves the base point to the origin, and then composing with $(Df(x_0))^{-1}$ on the left normalizes the derivative to the identity. If f^{-1} exists for this normalized version, it exists for the original f as well.

Define $g : E \rightarrow \mathbb{R}^n$ by $g(x) = f(x) - x$. Then $g(0) = 0$ and $Dg(0) = Df(0) - I = 0$. Since f is continuously differentiable, Dg is continuous, so there exists $r > 0$ such that

$$\|Dg(x)\| \leq \frac{1}{2} \quad \text{for all } x \in B(0, r).$$

By the mean value theorem inequality, for all $x, y \in B(0, r)$,

$$\|g(x) - g(y)\| \leq \frac{1}{2} \|x - y\|.$$

That is, g is a contraction on $B(0, r)$.

Existence of a local inverse. Fix any $y \in B(0, r/2)$ and define $T_y : \overline{B(0, r)} \rightarrow \mathbb{R}^n$ by $T_y(x) = y - g(x) = y - f(x) + x$. We check that T_y maps $\overline{B(0, r)}$ into itself: for $x \in \overline{B(0, r)}$,

$$\|T_y(x)\| \leq \|y\| + \|g(x)\| \leq \frac{r}{2} + \frac{1}{2} \|x\| \leq r.$$

Moreover T_y is a contraction since

$$\|T_y(x_1) - T_y(x_2)\| = \|g(x_1) - g(x_2)\| \leq \frac{1}{2} \|x_1 - x_2\|.$$

Since $\overline{B(0, r)}$ is a complete metric space, the Banach Fixed Point Theorem (Theorem 3.11) gives a unique fixed point $x \in \overline{B(0, r)}$ satisfying $T_y(x) = x$, i.e., $f(x) = y$. This defines the local inverse $f^{-1} : B(0, r/2) \rightarrow B(0, r)$.

Differentiability of f^{-1} . Let $y, y + k \in B(0, r/2)$ with corresponding preimages $x = f^{-1}(y)$ and $x + h = f^{-1}(y + k)$, so $k = f(x + h) - f(x)$. Since $Df(x) = I + Dg(x)$ is invertible for $x \in B(0, r)$ (as $\|Dg(x)\| \leq 1/2 < 1$, so $I + Dg(x)$ is invertible by Lemma 3.13), we have

$$k = Df(x)h + o(\|h\|), \quad \text{so} \quad h = (Df(x))^{-1}k + o(\|k\|),$$

where we used that $\|h\| \leq 2\|k\|$ from the contraction bound. Hence f^{-1} is differentiable at y with

$$D(f^{-1})(y) = (Df(f^{-1}(y)))^{-1}. \quad \square$$

4.3. Generalization to Banach Spaces. The proof of the $\mathbb{R}^n$ case relied only on the Banach Fixed Point Theorem and the mean value inequality - neither of which is specific to $\mathbb{R}^n$. So it is quite natural that the Inverse Function Theorem generalizes to maps between Banach spaces, which is the setting we actually work in when we apply it to systems of ODEs.

The generalization is as follows [T]: let X and Y be Banach spaces, $U \subseteq X$ open, and $F : U \rightarrow Y$ continuously Fréchet differentiable. If $DF(x_0) : X \rightarrow Y$ is a bounded linear isomorphism at some $x_0 \in U$, then there exist open sets $\mathcal{U} \subset U$ and $\mathcal{V} \subset Y$ with $x_0 \in \mathcal{U}$ and $F(x_0) \in \mathcal{V}$ such that $F : \mathcal{U} \rightarrow \mathcal{V}$ is a C^1 diffeomorphism with

$$D(F^{-1})(F(x_0)) = (DF(x_0))^{-1}.$$

The key point to note is that this theorem is inherently *local* — it guarantees an inverse only on some neighborhood $\mathcal{U}$ of x_0 , not on all of U . In general this locality cannot be avoided. However, as we will see in Section 5, when F is a linear map the Fréchet derivative satisfies $DF(x) = F$ for every x by Proposition 3.18, so the isomorphism condition holds everywhere and the local result becomes global. This is the mechanism that makes the IFT approach work for linear systems.

5. HOW IFT IS USED TO SOLVE LINEAR SYSTEMS OF FIRST-ORDER ODES

The classical methods of Section 2 all answer the same question: given the system $x'(t) = Ax(t)$, $x(0) = x_0$, how do we find the solution? In this section we ask a different question first: does a solution have to exist, and if so, is it unique? The Inverse Function Theorem answers both questions simultaneously by reformulating the problem as an inversion problem for a bounded linear operator between Banach spaces.

5.1. Reformulation of the problem. Rather than solving the initial value problem directly, we reformulate it as a single equation in a function space. We define the operator

$$(2) \quad F : C^1([0, T], \mathbb{R}^n) \longrightarrow C([0, T], \mathbb{R}^n) \times \mathbb{R}^n, \quad F(x) = (x'(t) - Ax(t), x(0)).$$

The domain $C^1([0, T], \mathbb{R}^n)$ is the space of continuously differentiable functions on $[0, T]$, and the codomain $C([0, T], \mathbb{R}^n) \times \mathbb{R}^n$ pairs a continuous function with a vector in $\mathbb{R}^n$. Both are Banach spaces under the norms established in Section 3.

The key observation is that solving

$$x'(t) = Ax(t), \quad x(0) = x_0$$

is exactly the same as finding $x \in C^1([0, T], \mathbb{R}^n)$ such that

$$F(x) = (0, x_0).$$

The first component $x'(t) - Ax(t) = 0$ encodes the differential equation, and the second component $x(0) = x_0$ encodes the initial condition. Rather than solving this directly, we ask: is F invertible? If so, the solution is simply

$$x(t) = F^{-1}(0, x_0),$$

and existence, uniqueness, and smooth dependence on x_0 all follow from the properties of F^{-1} .

5.2. Analysis on the operator. We now examine the structure of F . The first thing to notice is that F is linear - this is a structural fact which is *key* to the entire argument

Proposition 5.1. *The operator F defined in (2) is a bounded linear map, and $DF(x) = F$ for every $x \in C^1([0, T], \mathbb{R}^n)$ which follows from (Proposition 3.18).*

Proof. Linearity. For any $x, h \in C^1([0, T], \mathbb{R}^n)$ and $\alpha \in \mathbb{R}$,

$$\begin{aligned} F(x + \alpha h) &= ((x + \alpha h)'(t) - A(x + \alpha h)(t), (x + \alpha h)(0)) \\ &= (x'(t) - Ax(t) + \alpha(h'(t) - Ah(t)), x(0) + \alpha h(0)) \\ &= F(x) + \alpha F(h). \end{aligned}$$

Boundedness. For any $x \in C^1([0, T], \mathbb{R}^n)$,

$$\begin{aligned} \|F(x)\|_Y &= \|x' - Ax\|_\infty + |x(0)| \\ &\leq \|x'\|_\infty + \|A\| \|x\|_\infty + \|x\|_\infty \\ &\leq (2 + \|A\|) \|x\|_{C^1}, \end{aligned}$$

so F is bounded with $\|F\|_{\text{op}} \leq 2 + \|A\|$.

Fréchet derivative. Since F is linear, Proposition 3.18 gives $DF(x) = F$ for every $x \in C^1([0, T], \mathbb{R}^n)$ immediately. $\square$

The consequence of this is significant. Recall from Section 4 that the Banach space IFT requires $DF(x_0)$ to be a bounded linear isomorphism at some base point x_0 , and delivers a local inverse near that point. Here, $DF(x) = F$ at every x — the base point drops out entirely. This means if we can verify that F is a bounded linear isomorphism, the IFT will deliver a globally defined inverse, not just a local one. We next work through showing that it is a bounded linear isomorphism.

Lemma 5.2 (Picard–Lindelöf, $[\mathbf{M}; \mathbf{Y}]$). *Let $A \in \mathbb{R}^{n \times n}$ and $f \in C([0, T], \mathbb{R}^n)$. Then for any $v \in \mathbb{R}^n$, the initial value problem*

$$x'(t) = Ax(t) + f(t), \quad x(0) = v$$

has a unique solution $x \in C^1([0, T], \mathbb{R}^n)$.

Theorem 5.3. *The operator $F : C^1([0, T], \mathbb{R}^n) \rightarrow C([0, T], \mathbb{R}^n) \times \mathbb{R}^n$ is a bounded linear isomorphism.*

Proof. Boundedness was established above. We verify bijectivity; boundedness of F^{-1} then follows from the Open Mapping Theorem (Theorem 3.14)

Injectivity. Suppose $F(h) = (0, 0)$. Then $h'(t) = Ah(t)$ for all $t \in [0, T]$ and $h(0) = 0$. Set $\phi(t) = \|h(t)\|^2$. Then $\phi(0) = 0$ and

$$\phi'(t) = 2\langle h(t), h'(t) \rangle = 2\langle h(t), Ah(t) \rangle \leq 2\|A\| \|h(t)\|^2 = 2\|A\| \phi(t).$$

Applying Lemma 3.19 with $\alpha_0 = 0$ and $\beta(s) = 2\|A\|$ gives $\phi(t) \leq 0$ for all $t \in [0, T]$, hence $h = 0$. Alternatively, injectivity also follows from Lemma 5.2 applied with $f = 0$ and $v = 0$: the zero function is the unique solution to $h' = Ah$, $h(0) = 0$.

Surjectivity. Given any $(f, v) \in C([0, T], \mathbb{R}^n) \times \mathbb{R}^n$, we need $x \in C^1([0, T], \mathbb{R}^n)$ satisfying $x'(t) = Ax(t) + f(t)$ and $x(0) = v$. By Lemma 5.2, such an x exists and is unique.

We showed that F is both injective and surjective, so it is a *bijection*.

Bounded inverse. Since F is a bounded linear bijection between Banach spaces, the Open Mapping Theorem guarantees that F^{-1} is bounded. $\square$

5.3. The Key Idea. We have now verified everything required. The operator F is linear, its Fréchet derivative satisfies $DF(x) = F$ everywhere, and F is a bounded linear isomorphism. The Inverse Function Theorem can now be applied.

Since $DF(x) = F$ is a bounded linear isomorphism at every $x \in C^1([0, T], \mathbb{R}^n)$, the Banach space IFT (Section 4) guarantees a globally defined C^1 inverse

$$F^{-1} : C([0, T], \mathbb{R}^n) \times \mathbb{R}^n \longrightarrow C^1([0, T], \mathbb{R}^n).$$

The unique solution to the initial value problem $x'(t) = Ax(t)$, $x(0) = x_0$ is therefore

$$x(t) = F^{-1}(0, x_0).$$

Existence, uniqueness, and smooth dependence on x_0 all follow simultaneously.

A key observation to make here is that linearity is what has helped us out significantly. Generally, IFT produces an inverse for us only near a base point, but here since F is linear, $DF(x) = F$ holds everywhere and the isomorphism is global! This is what gives us local result to something global.

5.4. The Nonlinear Case. Now we briefly discuss what happens when we try to apply the IFT on a nonlinear ODE system

A nonlinear ODE system is one typically of the form

$$x'(t) = G(x(t)), \quad x(0) = x_0,$$

where $G : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is C^1 but not necessarily linear. The short answer is: the IFT still applies, but only locally.

We will apply the IFT and see what we get as a result.

We first define the *nonlinear* operator

$$\mathcal{F}(x) = (x'(t) - G(x(t)), x(0))$$

A direct computation shows that $\mathcal{F}$ is Fréchet differentiable, with

$$D\mathcal{F}(x) \cdot h = (h'(t) - DG(x(t))h(t), h(0)),$$

where $DG(x(t))$ is the Jacobian of G evaluated along x . This is no longer independent of x - the derivative changes as the base point changes, unlike the linear case where $DF(x) = F$ everywhere

We fix a base point x^* and suppose that $D\mathcal{F}(x^*)$ is a bounded linear isomorphism. This holds when $DG(x^*(t))$ is non-degenerate for all t , by the same injectivity and surjectivity arguments as we showed for the linear case.

The IFT then guarantees a local C^1 inverse near x^* : for initial conditions x_0 sufficiently close to $x^*(0)$, the IVP has a unique local solution depending smoothly on x_0 .

Example 5.4. Consider the system in $\mathbb{R}^2$

$$x'_1 = x_2, \quad x'_2 = -\sin(x_1),$$

which is the nonlinear pendulum. The Jacobian of the right-hand side is

$$DG(x) = \begin{pmatrix} 0 & 1 \\ -\cos(x_1) & 0 \end{pmatrix}.$$

At the equilibrium $x^* = (0, 0)^\top$, $DG(x^*) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, which is invertible. The IFT therefore guarantees a unique local solution near $x^* = (0, 0)^\top$ for initial conditions sufficiently close to the origin.

The argument cannot be extended globally beyond this step for three reasons:

- (1) **Finite-time blowup.** We may not have solutions at some finite time. The scalar equation $x' = x^2$, $x(0) = 1$ has the unique solution $x(t) = 1/(1 - t)$, which blows up as $t \rightarrow 1^-$.
- (2) **Non-uniqueness.** Without a Lipschitz condition on G , solutions may not be unique. The equation $x' = 3x^{2/3}$, $x(0) = 0$ has both $x = 0$ and $x(t) = t^3$ as solutions.
- (3) **Loss of the isomorphism condition.** Even when a smooth global solution exists, $D\mathcal{F}(x)$ may fail to be an isomorphism away from x^* - for instance if $DG(x(t))$ becomes degenerate along the solution. The IFT cannot be applied at such points.

In the linear case, none of these failures can occur. Linearity forces $DF(x) = F$ everywhere, the isomorphism condition is global, and the argument goes through without restriction. The nonlinear case shows that this is a genuinely special feature of linear systems, not a general fact.

6. EXTENSION TO NON-AUTONOMOUS SYSTEMS

The framework that we have seen and set up so far, can be extended quite naturally to the *linear* non-autonomous case where we have the system

$$(3) \quad x'(t) = A(t)x(t) + f(t), \quad x(0) = x_0,$$

where $A : [0, T] \rightarrow \mathbb{R}^{n \times n}$ is continuous and $f \in C([0, T], \mathbb{R}^n)$

The operator F from Section 5 is replaced by

$$\tilde{F}(x) = (x'(t) - A(t)x(t) - f(t), x(0)),$$

which is affine.

Its linear part $\hat{F}(x) = (x'(t) - A(t)x(t), x(0))$ satisfies $D\tilde{F}(x) = \hat{F}$ for all x , by the same argument as before. The three checks - injectivity via Gronwall's inequality, surjectivity via the variation of parameters formula from Section 2, and bounded inverse via the Open Mapping Theorem - carry through with $A(t)$ in place of A , and the IFT again delivers a globally defined inverse. This argument can be justified using the tools we have already established.

7. ACKNOWLEDGEMENTS

I would like to thank Ben for being an excellent instructor and a wonderful semester of MA 442. The material covered in this course, particularly the functional-analytic tools were extremely insightful for me and is something I will take with me forever in my mathematical journey. Ben's lectures made it possible for me to approach this problem in ODE theory from a perspective that I would not have otherwise thought about. This was directly influenced by what was learned in class. I would also like to thank Dr. Aaron Yip for guiding me and giving me resources to reference for this project.

REFERENCES

- [1] W. E. Boyce and R. C. DiPrima, *Elementary Differential Equations and Boundary Value Problems*, 11th ed., Wiley, 2017. [BDP]
- [2] N. Yip, *MA 543 Lecture Notes*, Purdue University. [Y]
- [3] J. D. Meiss, *Differential Dynamical Systems*, Society for Industrial and Applied Mathematics, 2007. [M]
- [4] T. Tao, *Analysis I*, Hindustan Book Agency, 2006. [TI]
- [5] D. B. McReynolds, *MA 442, Spring 2026: Definition Sheet*, January 27, 2026. [D]
- [6] T. Tao, *Analysis II*, Hindustan Book Agency, 2006. [TII]
- [7] S. van Straten, *Multivariable Calculus*, Imperial College London. Available at: <https://ma.imperial.ac.uk/~svanstri/Files/multivariable-calculus.pdf>. [VS]
- [8] Wikipedia, *Inverse Function Theorem*. Available at: https://en.wikipedia.org/wiki/Inverse_function_theorem [W].
- [9] G. Teschl, *Ordinary Differential Equations and Dynamical Systems*, American Mathematical Society. Available at: <https://mat.univie.ac.at/~gerald/ftp/book-ode/ode.pdf>. [T]